



IK Avatar Driver
V 1.0.3

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Please consider leaving a review to let me know how I'm doing, it means a lot to me.

The asset is still being improved, if you have any feedback please join the discussion forum:
<https://forum.unity.com/threads/ik-avatar-driver.811539/>

Introduction

IK Avatar Driver is an arcade style car controller with a procedurally animated IK avatar driver, you can customize it with your own car and humanoid-character models. The avatar is modular and can be used with Simple Traffic System, Edy's Vehicle Physics, NWH Vehicle Physics and Realistic Car Controller.

Car Controller

This section will cover the car controller prefabs and their available options. Drag & drop one of these prefabs into an empty scene to use it. They are located at: **Assets\TurnTheGameOn\IKAvatarDriver\Prefabs**

CarController_DriverLeft

- Avatar setup to drive on the left side of the car model.

CarController_DriverRight

- The avatar setup to drive on the left side of the car model.

CarController_DriverLeft_Mobile

- Avatar setup to drive on the left side of the car model.
- Prefab variant with mobile options preset.

CarController_DriverRight_Mobile

- The avatar setup to drive on the left side of the car model.
- Prefab variant with mobile options preset.

Car Controller Scriptable Object Profiles

These prefabs have lots of customization options and use scriptable object profiles to manage these settings. The root of the prefab holds all script components with references to these profiles. The profiles are located at: **Assets\TurnTheGameOn\IKAvatarDriver\Data**

CarDriveSettings

- Modify settings to change vehicle behavior.

CarInputSettings

- Controls the car input parameters.

CarWheelSettings

- Enable/disable skid audio and effect.

Avatar Driver

This prefab is already set up on the car controller prefabs. You can use this prefab with **Edy's Vehicle Physics**, **NWH Vehicle Physics** and **Realistic Car Controller**. They are located at:

Assets\TurnTheGameOn\IKAvatarDriver\Prefabs

Avatar Driver Settings Scriptable Object Profiles

The avatar driver prefab uses scriptable object profiles to manage most of the options. Each car controller uses a different profile with settings configured for the car controller prefab's requirements. They are located at:

Assets\TurnTheGameOn\IKAvatarDriver\Data

AvatarDriver_EVP - Edy's Vehicle Physics

- Video Tutorial: <https://youtu.be/UMmnO66KXwc>
- Preconfigured to use with the Edy's Vehicle Physics asset store package. To use this prefab, you need to import the **AvatarDriver_EVP** support package located at:
Assets\TurnTheGameOn\IKAvatarDriver\3rdPartySupport
- Open the Edy's Vehicle Physics **The City - Simple Scene - Drift** demo scene.
- Drag & drop the **AvatarDriver_EVP** prefab as a child of the **Sport Coupe Drift** prefab.
- Set the **AvatarDriver_EVP** transform position to x -0.027 y -0.088 z 0.773
- Expand the **AvatarDriver_EVP** object to reveal its children, expand the **BodyJoint** object to reveal it's child **Driver** and assign the root **Sport Coupe Drift** to the **Vehicle Rigidbody** field of the **Avatar Driver** script.

AvatarDriver_NWH - NWH Vehicle Physics 2

- Video Tutorial: <https://youtu.be/XHI36ejUUTQ>
- Preconfigured to use with the NWH Vehicle Physics asset store package. To use this prefab, you need to import the **AvatarDriver_NWH** support package located at:
Assets\TurnTheGameOn\IKAvatarDriver\3rdPartySupport
- Open the NWH Vehicle Physics **Racetrack** demo scene.
- Drag & drop the **AvatarDriver_NWH** prefab as a child of the **Sports Car** prefab.
- Unpack the **AvatarDriver_NWH** prefab completely.
- Set the **AvatarDriver_NWH** transform position to x 0.034 y -0.252 z 0.611
- Expand the **AvatarDriver_NWH** object to reveal its children, expand the **BodyJoint** object to reveal it's child **Driver**.
- Assign the **Steering Wheel** that's a child of the **Sports Car** to the **Read Only Steering Wheel** field of the **Avatar Driver** script.
- Assign the root **Sports Car** to the **Vehicle Rigidbody** field of the **Avatar Driver** script.
- Assign the root **Sports Car** to the **Vehicle Controller Reference** field of the **AvatarDriver_NWHReference** script.
- Move the **IK Steering Wheel** from the **Avatar Driver** object to be a child of the **Steering Wheel** object. Set the position x and y values to 0, adjust z position as needed. You can also scale the
- Expand the **IK Steering Wheel** object and disable or remove the **Steering Wheel Mesh** object. Hint: you can use this object as a guide to help position your steering wheel for different vehicles. **IK Steering Wheel** object to adjust the IK Hand Targets if needed.

AvatarDriver_RCC - Realistic Car Controller

- Video Tutorial: <https://youtu.be/mvIOB8mw0zA>
- Preconfigured to use with the Realistic Car Controller asset store package. To use this prefab, you need to import the **AvatarDriver_RCC** support package located at:
Assets\TurnTheGameOn\IKAvatarDriver\3rdPartySupport
- Open the RCC demo scene named **RCC City** to start.
- Disable the default scene car **M3_E36**.
- Add the **Model_Misc_Buggy** car prefab to the scene, this is a good test prefab since it exposes the interior and the avatar driver can be easily seen.
- Drag & drop the **AvatarDriver_RCC** prefab as a child of the **Model_Misc_Buggy** prefab.
- Set the **AvatarDriver_RCC** transform position to x 0.034 y -0.835 z 0.161
- Set the **AvatarDriver_RCC** transform scale to x 0.58 y 0.58 z 0.58
- Expand the **AvatarDriver_RCC** object to reveal its children, expand the **BodyJoint** object to reveal its child **Driver** and assign the root **Model_Misc_Buggy** to the **Vehicle Rigidbody** field of the **Avatar Driver** script.
- Find the **Gear Label** object from the **RCCCanvas** object and assign it to the **Gear Text** field of the **Avatar Driver** script.
- Assign the root **Model_Misc_Buggy** to the **Car Controller Reference** field on the **Avatar Driver_RCC Reference** script.

AvatarDriver_STS - Simple Traffic System

- Video Tutorial: https://youtu.be/_HaOGVF9wKk
- Preconfigured to use with the Simple Traffic System asset store package. To use this prefab, you need to import the **AvatarDriver_STS** support package located at:
Assets\TurnTheGameOn\IKAvatarDriver\3rdPartySupport
- Open the STS demo scene named **Demo** to start.
- Open the **AITrafficCar__Random** AI car prefab, this is a good test prefab since it's used on every car in this demo scene. If the performance cost is too heavy, reduce the number of cars in the scene.
- Drag & drop the **AvatarDriver_STS** prefab as a child of the **AITrafficCar__Random** prefab and move it into place.
- This default car prefab does not have an interior, so you will need a better car model.
- Expand the **AvatarDriver_RCC** object to reveal its children, expand the **BodyJoint** object to reveal its child **Driver** and assign the root **AITrafficCar__Random** to the **Vehicle Rigidbody** field of the **Avatar Driver** script.
- Assign the root **AITrafficCar__Random** to the **Car Controller Reference** field on the **Avatar Driver_STS Reference** script.

Replacing The Avatar Driver Model

- Video Tutorial: <https://youtu.be/mcFFZVaU3fQ>
- Import the example character model:
<https://assetstore.unity.com/packages/3d/characters/robots/space-robot-kyle-4696>
- Select the Robot Kyle fbx and change the Rig Animation Type to Humanoid.
- The avatar model can be replaced with any humanoid model.
- Open the Demo scene, select the Player_Car_LeftSteering, right click it and select Unpack Prefab Completely.
- Expand the AvatarDriver object, expand the BodyJoint object, expand the RaceDriver object - we'll replace the children of the race driver object with the required objects of the new avatar model. Note: You can perform these steps directly on any of the AvatarDriver prefabs.
- Drag and drop the Robot Kyle fbx as a child of the Race Driver object, right click Robot Kyle and select Unpack Prefab Completely.
- Expand the Robot Kyle object and select the children (Robot 2 and Root) objects, move these objects to be children of the Race Driver instead of Robot Kyle.
- If you're using the IK Driver Vehicle Controller and camera systems, you'll want to find the camera that's attached to the default avatar and bring that over to your new avatar rig. Expand Hips, Spine, Spine1, Spine2, Neck, Head and select HelmetCameraShakeTransform - you can copy the transform values for a rough approximation, but this object will likely need to be adjusted a bit to properly line up with the new avatar's head to account for any offset differences between the avatar models. Move this to be a child of the Robot Kyle Head bone and paste the copied transform values (for this model, add -90 to the x rotation value and +90 to the z rotation value).
- Select the original character model (Helmet, Hips, Suit) objects and delete them.
- Select the unused Robot Kyle object and delete it.
- Select the Race Driver object and assign the Robot Kyle Avatar object from the fbx to the Animator.
- Press Play and Robot Kyle will now be driving instead of the default driver.